Markov Decision Process (MDP)

S : state space $\quad s_t \in S$

A : action space $\quad a_t = \pi(s_t)$

$P^{s_{t+1}}_{s_t \pi} = p(s_{t+1} \mid s_t, \pi)$: state transition probabilities

$\gamma \in [0,1]$: discount factor $\quad R$: reward function

$R(a_t, s_t) \in A \times S \to \mathbb{R}$ or $R(s_t) \in S \to \mathbb{R}$

$$s_t, a_t \to s_{t+1}, a_{t+1} \to s_{t+2}, a_{t+2} \to \dots \to s_\infty, a_\infty$$

① $s_{t+1} \sim P^{s_{t+1}}_{s_t \pi} = p(s_{t+1} \mid s_t, a_t)$

② $a_t = \pi(s_t) \qquad s_t \to \boxed{\pi} \longrightarrow a_t$

$$G_t = R(s_t, a_t) + \gamma R(s_{t+1}, a_{t+1}) + \gamma^2 R(s_{t+2}, a_{t+2}) + \dots + R(s_\infty, a_\infty)$$

③

$$V^\pi(s_t, a_t) = E\left[G_t \mid s_t, a_t\right]$$

$$= E\left[R(s_t, a_t) + \gamma R(\underline{s_{t+1}}, a_{t+1}) + \gamma^2 R(\underline{s_{t+2}}, a_{t+2}) \dots \mid s_t, a_t\right]$$

$$\Downarrow$$

Bellman equation:

$$V^\pi(s_t, a_t) = R(s_t, a_t) + \gamma \sum_{s_{t+1} \in S} P^{s_{t+1}}_{s_t a_t} \underline{V^\pi(s_{t+1}, a_{t+1})} \leftarrow$$

$$E\left[R(\underline{s_t, a_t}) + \gamma R(s_{t+1}, a_{t+1}) + \gamma^2 R(s_{t+2}, a_{t+2}) \dots \,\middle|\, \underline{s_t, a_t}\right]$$

$$R(s_t, a_t) + \underbrace{E\left[\gamma R(\underline{s_{t+1}}, \underline{a_{t+1}}) + \gamma^2 R(s_{t+2}, a_{t+2}) \dots\right]}$$

$$\gamma \underbrace{E\left[R(s_{t+1}, a_{t+1}) + \gamma R(s_{t+2}, a_{t+2}) + \dots \,\middle|\, s_t, a_t\right]}_{V^\pi(s_{t+1}, a_{t+1})}$$

$$E\left[V^\pi\right] = \sum P^{s_{t+1}}_{s_t, a_t} V^\pi(s_{t+1}, a_{t+1})$$

$$P^{s_{t+1}}_{s_t} = \sum_{a_t \in A} P^{s+1}_{s_t, a_t} \pi(s_t)$$

$$= \sum_{a_t \in A} p(s_{t+1} | s_t, a_t) \, \pi(a_t | s_t)$$

$$= p(s_{t+1} | s_t)$$

$$V^\pi(s_t, a_t) = \underbrace{R(s_t, a_t)}_{①} + \underbrace{\gamma \sum_{s_{t+1} \in S} P^{s_{t+1}}_{s_t, a_t} V^\pi(s_{t+1}, a_{t+1})}_{②}$$

$$\underline{V^\pi(s_t, a_t)} = \left(I - \gamma P^{s_{t+1}}_{s_t, a_t}\right)^{-1} V^\pi(s_t, a_t)$$

Value Iteration / Policy Evaluation:

∅	0	0
D	0	0
0	0	∅

Algorithm:

① For each state $s_t \in S$, randomly initialize $V(s_t)$

② Repeat until convergence {

For every s_t, update $V(s_t, a_t) := R(s_t, a_t)$

Policy Iteration

$$+ \boxed{\max_{a=\pi(s_t)} \gamma \sum_{s_{t+1}} P_{s_t, a_t}^{s_{t+1}} V(s_{t+1}, a_{t+1})}$$

}

Value Iteration (action is implicit (hidden)):

$$V(s_t) := R(s_t) + \gamma \sum_{s_{t+1}} P_{s_t}^{s_{t+1}} V(s_{t+1})$$

Policy Evaluation (a_t chosen randomly):

$$V(s_t, a_t) := R(s_t, a_t) + \gamma \sum_{s_{t+1}} P_{s_t, a_t}^{s_{t+1}} V(s_{t+1}, a_{t+1})$$

Policy Iteration (a_t is arg max of 2nd term).

$$V(s_t, a_t) := R(s_t, a_t) + \gamma \max_{a_t} \sum_{s_{t+1}} P_{s_t, a_t}^{s_{t+1}} V(s_{t+1}, a_{t+1})$$